Where to Measure Next? Evaluating Feedback-Driven Region Selection for Spatial Transcriptomics

Anonymous authors
Paper under double-blind review

Abstract

Spatial transcriptomics is costly, so many studies profile only selected regions of a section, and recent methods choose each next region from expression already measured. A recent benchmark of active spot selection for training predictors across patients reported inconsistent gains over random sampling; we ask instead whether, within a slide, measured feedback improves selection over a fixed design matched to the objective. We built a retrospective simulator from 22 human HEST-1k slides (18 studies, four tissues), fixed the image encoder and imputer, registered three analysis stages in timestamped commits before running them, and drew inference at the study level. Each feedback policy was compared with a non-adaptive design chosen on held-out studies and with itself fed uninformative values. **RQ1:** for whole-section reconstruction, a policy targeting the imputer's error was below the fixed design at every budget (0.5–20%), significantly at four of six and by up to 0.092 Pearson correlation. **RQ2:** for capturing heterogeneous regions, informative feedback raised recall over permuted feedback by up to 0.063 without surviving multiplicity correction, and a diversity sampler that uses no feedback did about as well. **RQ3:** spending half or more of the budget on a fixed design and the rest on feedback never beat that design; from 2% upward every hybrid was within 0.007 correlation and 0.006 recall of it, a null with at least 80% power to detect gains of 0.02 and 0.03 at 2–10%. **RQ4** (exploratory): batches of five neither eroded nor established a detection advantage. No feedback policy we tested outperformed a well-chosen fixed design on either objective, within simulated spot-level budgets, one imputer, and computational endpoints. Adaptive acquisition methods should be benchmarked against objective-matched fixed designs and against themselves without informative feedback, not only against random selection.

1 Introduction

Spatial transcriptomics measures gene expression without dissociating tissue, so expression can be read in its histological context (Ståhl et al., 2016; Rao et al., 2021; Moses & Pachter, 2022). Whole-transcriptome assays remain costly, and region-of-interest platforms such as GeoMx DSP (Merritt et al., 2020) profile only selected regions of a section. Expression elsewhere can be predicted from the co-registered hematoxylin and eosin (H&E) image (Xie et al., 2023; Jaume et al., 2024). Which regions are measured therefore determines both how well the unmeasured tissue can be reconstructed and whether distinct tissue compartments are measured at all.

Choosing regions is an experimental-design problem, and recent methods solve it sequentially. SOFisher (Li et al., 2026) uses reinforcement learning to move fields of view toward regions enriched for a target signal, and SCR^2-ST (Zhu et al., 2026a) learns where to sample spots to build training sets for expression prediction. S2-omics (Yuan et al., 2025) takes the opposite approach and selects regions from histology alone, before any measurement. The appeal of sequential selection follows active learning (Settles, 2009): each measurement can inform the next. Its cost is a measurement turnaround between choices. A recent benchmark of active spot selection for training expression predictors found inconsistent gains over random sampling at small budgets (Zhu et al., 2026b), but addressed a different decision (Section 2). We are not aware of an evaluation

that compares feedback-driven region selection within a slide against fixed designs, whose selections do not depend on measured values, matched to the objective while ablating the measured feedback itself.

Theory gives different expectations for different goals. For reconstructing a Gaussian-process field with a known kernel, posterior variance does not depend on the observed values, so a design fixed before any measurement is as good as a sequential one; when the model's parameters are uncertain, sequential designs can do better (Krause et al., 2008). Imputers learned from images are not Gaussian processes with known kernels, so for reconstruction the question is empirical. Finding regions defined by their expression is closer to active search, in which each choice exploits the labels already observed to locate further targets (Garnett et al., 2012).

Beating random selection does not show that feedback helps, because a policy can win through its selection rule alone. An evaluation that can separate these cases needs four features. The baseline must be a design that could be chosen before the target slide is profiled, because selecting the best design on the evaluation data biases the comparison (Varma & Simon, 2006; Cawley & Talbot, 2010). Generalisation must be assessed across studies, not within them (Bernau et al., 2014), with slides from one study treated as dependent (Field & Welsh, 2007). The contribution of feedback must be isolated by giving the same policy uninformative values, in the spirit of permutation tests for learning algorithms (Ojala & Garriga, 2010). Finally, null results need positive controls and power analysis to be interpretable (Altman & Bland, 1995).

We built such an evaluation and asked four questions. **RQ1**: does conditioning region selection on measured expression improve whole-section reconstruction over a feasible non-adaptive design? **RQ2**: does it improve the capture of molecularly heterogeneous regions? **RQ3**: does feedback add to a strong fixed design when it allocates only the remainder of the budget? **RQ4**, an exploratory stress test: does a detection advantage persist when measurements arrive in batches? The answer held across three registered stages: no tested feedback policy outperformed a well-chosen fixed design, and for reconstruction the fixed design was significantly better. We contribute (i) a reproducible simulator on 22 human slides from 18 studies with feasible, objective-matched baselines, feedback ablations, and study-level inference; (ii) a registered negative result for both objectives, made interpretable by positive controls and minimum detectable effects; and (iii) an account of the result: the imputer's uncertainty depends only on where regions are profiled, so feedback can aid reconstruction only through its predictions, and for detection a diversity sampler without feedback finds the heterogeneous regions about as well.

2 Related work

Expression from histology. Expression has been predicted from H&E by contrastive retrieval (Xie et al., 2023), flow matching over spatial neighbourhoods (Huang et al., 2025), diffusion (Zhu et al., 2025), and autoregressive generation (Ouyang et al., 2025), and benchmarked on HEST-1k (Jaume et al., 2024). These methods take the measured regions as given. We hold the predictor fixed and vary only the regions it receives.

Choosing what to measure. Pool-based active learning selects queries by uncertainty (Lewis & Gale, 1994), expected error reduction (Roy & McCallum, 2001), or coverage of the input space (Sener & Savarese, 2018). Sensor placement offers the matching non-adaptive designs: variance- and mutual-information-greedy selection, which are near-optimal for Gaussian-process reconstruction (Krause et al., 2008). Adaptive submodularity extends such guarantees to greedy policies that condition on observations (Golovin & Krause, 2011), and active search addresses the related goal of collecting positives (Garnett et al., 2012). In batch acquisition, the members of a batch are chosen without each other's labels (Kirsch et al., 2019), and global optimisation commonly begins with a space-filling design before switching to adaptive queries (Jones et al., 1998). Active feature acquisition casts measurement as a sequential policy (Yin et al., 2020).

Design in spatial omics. Gene-panel selectors fix which genes to measure (Covert et al., 2023; Lu et al., 2026). Region selectors range from image-based rankers such as SpatialFinder, which score regions from H&E before any expression is measured (Xu et al., 2026), and histology-based heterogeneity selection (Yuan et al., 2025) to sequential reinforcement-learning samplers (Li et al., 2026; Zhu et al., 2026a). Our evaluation includes reimplementations of these objectives and a native run of SOFisher. The closest evaluation is a

retrospective benchmark of active spot selection on breast and skin cancer cohorts (Zhu et al., 2026b), which compared uncertainty- and diversity-based strategies with random sampling and found that active strategies did not consistently win at small budgets. Its question differs from ours in three ways. It selects spots to train a morphology-to-expression predictor that is evaluated on held-out patients, whereas we select regions within a slide for a fixed imputer. Its baseline is random sampling, whereas ours are non-adaptive designs chosen for each objective. It compares selection strategies, whereas we hold each strategy fixed and remove the information in the measured values.

3 Methods

3.1 Acquisition simulator

A slide is a set of regions, each with an image embedding, spatial coordinates, and measured expression. A policy reveals regions one at a time until a budget, a fixed fraction of the slide's regions, is spent; each revealed region exposes its measured expression to the policy. The simulator is retrospective, as in pool-based active-learning evaluation (Settles, 2009): every region has been measured, so any selection can be scored against ground truth. For RQ4, regions are revealed in rounds, and within a round the policy can use only measurements available at its start. Budgets range from 0.5% of regions (9 to 45 regions depending on slide size, comparable to ROI-based profiling) to 20%.

3.2 Slides and preprocessing

We used 22 human HEST-1k slides (Jaume et al., 2024) from prostate (6), kidney (4), lung (6), and colon (6), of which 7 are Visium HD (de Oliveira et al., 2025) and 15 standard Visium (Ståhl et al., 2016) (Appendix A). The tissues are the four used to evaluate the image-based ranker we compare against (Xu et al., 2026). Selection rules were fixed before any result on these slides: human tissue, whole-transcriptome assay, at least 1,000 spots under tissue, a source image of at most 0.6 μm/px so that image patches are not substantially upsampled, one slide per specimen, and at most two slides per study. The slides come from 18 studies, which serve as the independent units of inference.

A region is one native spot, the unit at which HEST provides both expression and an image patch. Visium spots are 55 μm wide on a 100 μm pitch. HEST distributes Visium HD data as 128 μm square pseudo-spots pooled from 16 μm bins, the default of its Visium HD reader,[1] so an HD region covers about seven times the area of a Visium spot. Slides contain 1,766 to 9,080 regions. Each region's image input is the HEST 224 px patch at 0.5 μm/px (Jaume et al., 2024), embedded with UNI2-h, a later release of the pathology foundation model UNI (Chen et al., 2024),[2] used frozen. Following the HEST benchmark, expression is log1p-transformed and genes detected in fewer than 10% of regions are removed (Jaume et al., 2024).

3.3 Objectives

Reconstruction (RQ1, RQ3): a fixed imputer predicts the unprofiled regions, scored by the per-gene Pearson correlation over those regions averaged over the 50 most variable genes (PCC), the HEST benchmark target (Jaume et al., 2024). *Detection* (RQ2–RQ4): regions of interest (ROIs) are the 20% of regions with the highest local molecular heterogeneity, which we define as the variance of top-50-gene expression across each region's 8 nearest spatial neighbours. The definition is ours; it is motivated by the goal of capturing molecular heterogeneity that drives S2-omics (Yuan et al., 2025), which itself selects regions from histology. The ROI fraction equals the largest budget, so a perfect policy could capture every ROI. We score recall, the fraction of ROIs profiled; an oracle that profiles regions in order of true heterogeneity bounds it from above. Gene sets and ROIs are computed from ground truth and used only for scoring.

[1] https://hest.readthedocs.io/en/latest/generated/hest.readers.VisiumHDReader.html
[2] https://huggingface.co/MahmoodLab/UNI2-h

3.4 Imputer

One imputer serves every policy, so differences between policies reflect selection alone. It maps image principal components through random Fourier features (Rahimi & Recht, 2007), fits a ridge regression to the profiled regions, and propagates the resulting predictions over a fixed 8-nearest-neighbour graph in image and spatial features while holding profiled regions at their measured values, as in harmonic label propagation (Zhu et al., 2003). Its use of spatial neighbourhoods is inspired by predictors such as STFlow (Huang et al., 2025), but it is far simpler, which makes it cheap enough to refit after every measurement. Its predictions depend on the measured values. Its uncertainty, one minus the graph influence that reaches a region from profiled nodes, depends only on which regions are profiled. A k-nearest-neighbour imputer and a regression-kriging imputer (Hengl et al., 2007) were evaluated on an earlier slide set (Appendix C). Hyperparameters of the imputer and policies were fixed before the registered analysis in which each was first used and were not tuned on its outcomes; those of the imputer were set during development on one prostate slide (Appendix B).

3.5 Policies

A *fixed design* is one whose selections do not depend on measured expression; a *feedback policy* is one whose selections do.

Non-adaptive designs, fixed designs computed from images and coordinates alone. Random; farthest-point selection in space (grid), a greedy k-centre cover (Gonzalez, 1985); k-means (Lloyd, 1982) on image embeddings, taking the region nearest each centroid; a one-shot ranker after SpatialFinder (Xu et al., 2026), which scores regions once from image embeddings by rare-cluster membership and distance to the cluster centroid, a simplification of its diversity score without the cell classifier or pathologist labelling ("one-shot" below); and variance- and mutual-information (MI)-greedy placement (Krause et al., 2008) with an RBF kernel on image and coordinate features and a median-heuristic lengthscale (Garreau et al., 2017). Grid, k-means, and MI-greedy are the *coverage designs* of the stage-1 analyses.

Sequential fixed designs. Error-reduction (Roy & McCallum, 2001) picks the region whose 15 nearest unprofiled neighbours in the imputer's feature space carry the most uncertainty, and maximum-uncertainty picks the most uncertain region (Lewis & Gale, 1994). Because the imputer's uncertainty does not depend on measured values, both are fixed designs computed one region at a time; for error-reduction we confirmed that permuted values leave its selections unchanged.

Feedback policies. The *residual* policy keeps the error-reduction form but replaces the uncertainty with an estimate of the imputer's actual error. It computes the leave-one-out residual of the image ridge model at each profiled region from the hat matrix (Hastie et al., 2009), averages it over each unprofiled region's 10 nearest profiled neighbours, and multiplies by one minus the graph influence. The *local-variance* policy, our detection policy, scores a candidate by the kernel-weighted variance of measured expression around it plus 0.5 times its distance to the nearest profiled region, an exploitation term for heterogeneity balanced by an exploration term as in active search (Garnett et al., 2012). While no region has been measured, it selects the region nearest the slide centroid.

Competitor objectives. A diversity sampler applies farthest-point selection in joint image and spatial features, the coverage criterion of core-set active learning (Sener & Savarese, 2018). It approximates the coverage and diversity terms of the SCR^2-ST reward, which rewards covering single-cell states, cell-type diversity, and spatial spread (Zhu et al., 2026a). An adaptive ROI-seeking sampler follows the reward of SOFisher (Li et al., 2026). Both reimplement stated objectives, not the authors' trained models.

Hybrids (RQ3). A hybrid takes its first $\lceil fn \rceil$ regions, for a budget of n regions and $f \in \{0.5, 0.8\}$ (the 50% and 80% hybrids), from a non-adaptive *base design* and the rest from a feedback policy: the residual policy for reconstruction and local-variance for detection. The base design is the same for every slide at each budget and objective: k-means at 0.5–1% and MI-greedy at 2–20% for reconstruction, and variance-greedy at 2–10% and the one-shot ranker at 20% for detection, chosen from stage-2 results as described under Comparators. This mirrors the space-filling start of adaptive global optimisation (Jones et al., 1998). Because the base designs select in a predetermined order, the first phase is exactly the prefix of the design's own selection

(verified for all four designs at both fractions), so a hybrid and its base design differ only in how the last $1 - f$ of the budget is spent.

3.6 Feedback ablations

Each feedback policy and hybrid was also run with uninformative feedback while its selection rule was left unchanged. Under *permuted* feedback, profiling a region reveals the expression of a randomly chosen other region, so the residual policy's leave-one-out model, for example, is fitted to permuted values. Under *constant* feedback, every region appears to have the slide's mean expression. The imputer and the scores always use the true values. A policy that gains from feedback should outperform its own permuted version, the logic of permutation tests for whether a learner exploits real structure (Ojala & Garriga, 2010). Stage 2 used one permutation; stage 3 used two per configuration and repeated the stage-2 ablations with up to four further seeds.

3.7 Comparators

The primary stage-2 comparator is a *feasible* design chosen without the target slide. For each held-out study, the non-adaptive design (grid, k-means, variance-greedy, MI-greedy, or one-shot) with the best mean score on the other 17 studies at that budget and objective is applied to the held-out slides, a leave-one-study-out form of cross-study validation (Bernau et al., 2014). The *hindsight benchmark*, the best non-adaptive design for each slide chosen after seeing that slide's result, is reported only for reference, since selecting on the evaluation data inflates the comparator (Varma & Simon, 2006; Cawley & Talbot, 2010). Stage 3 compared each hybrid with its base design, the design chosen as feasible for most held-out studies at that budget and objective. The base designs were chosen on stage-2 results from the same slides, which favours the baseline.

3.8 Statistical analysis and registration

Each margin is a per-slide difference between two policies. We report its mean with a 95% percentile bootstrap interval (Efron & Tibshirani, 1993) that resamples whole studies (Field & Welsh, 2007) (5,000 resamples), and an exact two-sided sign-flip test over the 2^{18} study-level sign patterns (Winkler et al., 2014). Because resampling as few as 18 clusters can give intervals that are too narrow (Cameron et al., 2008), we also report, post hoc, intervals obtained by inverting the sign-flip test (Appendix M). Intervals are not adjusted for multiplicity; p_H denotes p-values adjusted by Holm's procedure within a registered family (Holm, 1979). To make null results interpretable (Altman & Bland, 1995), we report the minimum detectable effect (Bloom, 1995): the smallest true mean margin detected with 80% power (Cohen, 1988), estimated by simulation that resamples studies around the observed dispersion (Arnold et al., 2011). For a family of m margins we compute it at $\alpha = 0.05/m$ (MDE_H), a conservative bound for the Holm decision.

The sign-flip test assumes exchangeable study margins. This holds for a comparator that is the same for every study but not exactly for the feasible one, whose selection for each study uses the other 17. Following the principle that model selection must be repeated inside resampling (Varma & Simon, 2006), we checked the stage-2 results post hoc with a bootstrap that repeats the selection in every replicate and with MI-greedy as the comparator on every slide (Appendix F).

Analyses were registered in the project's version-controlled repository in three stages (Nosek et al., 2018), each before its experiments and after the previous stage's results were known.[3] Stage 1 compared policies with hindsight benchmarks in a family of 24 margins, including the first batched experiments and contiguous blocks of 7 regions around a chosen seed, which approximate field-of-view acquisition (Appendix D). Stage 2 introduced the feasible design, the residual policy, and the feedback ablations, with a primary family of 18 margins: residual minus feasible at six budgets (RQ1), and at four budgets local-variance minus its permuted version, minus the feasible design, and minus the diversity sampler (RQ2). Stage 3 tested the hybrids, with a primary family of 20 margins, each hybrid minus its base design run for the whole budget (RQ3), and a secondary family of 24 that compared hybrids with their permuted versions and repeated the batched

[3]Verbatim excerpts of the three plans and their commit dates are in the supplementary material (`registered_plans/`); the plans label RQ4 as Q3.

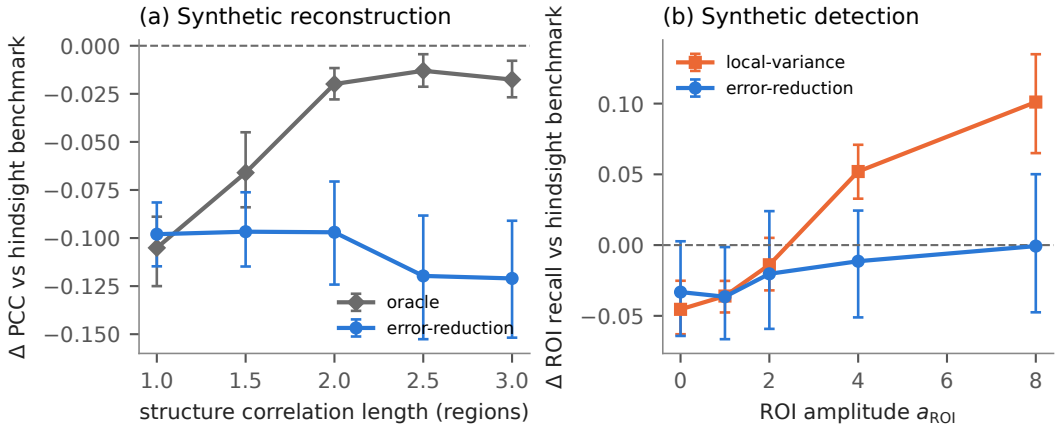


Figure 1: Synthetic controls (8 slides per point, 10% budget, mean and 95% bootstrap CI over slides). (a) Reconstruction margin over the hindsight benchmark of grid and k-means versus the correlation length of the hidden structure, for the oracle ordering by true local difficulty and for error-reduction. (b) Detection-recall margin over the benchmark of random, grid, and k-means versus ROI amplitude.

experiments with the base detection design and permuted controls (RQ4). In every stage a benefit was to be claimed only for a positive margin with $p_H < 0.05$. Results are labelled confirmatory (registered primary), secondary (registered, Holm within the secondary family), descriptive, or post hoc.

3.9 Positive controls

A null result is informative only if the protocol could have registered an effect. In synthetic slides of 20×20 regions with spatially clustered structure that the image cannot predict (8 slides per setting, 10% budget, k-NN imputer), we varied the correlation length of the hidden structure and the amplitude a_{ROI} with which ROIs appear in measured expression. Synthetic margins are taken over the hindsight benchmark of the simple designs: the better of grid and k-means on each slide, and for detection also random. On the 22 real slides we injected 20 genes carrying three spatial hotspots of radius 5% of the slide diagonal, in the manner of simulated spatially variable genes (Svensson et al., 2018), with standard deviation equal to a spike amplitude $a_{\mathrm{spike}} \in \{0, 1, 2\}$ times the median of the slide's top-50 genes. Because the image cannot predict the hotspots, only measured values reveal them. Before running this control we set its success criterion: the residual policy beats the feasible design with an interval excluding zero at $a_{\mathrm{spike}} = 2$.

4 Results

4.1 Positive controls

The simulator registered an effect of feedback on detection. In synthetic slides, the local-variance policy's recall margin over this benchmark rose with ROI amplitude, from -0.045 at $a_{\mathrm{ROI}} = 0$ to $+0.052$ (95% CI $+0.033$ to $+0.071$) at $a_{\mathrm{ROI}} = 4$ and $+0.101$ ($+0.065$ to $+0.135$) at $a_{\mathrm{ROI}} = 8$, positive on all 8 slides at both amplitudes (Figure 1b). Reconstruction behaved differently: even an oracle that orders regions by true local difficulty, which no policy could deploy, stayed below the benchmark at every correlation length, by 0.105 and 0.066 PCC at the two shortest and 0.013 to 0.020 at the longer ones (Figure 1a).

The semi-synthetic control failed its criterion. At $a_{\mathrm{spike}} = 1$ and 2, all 20 injected genes entered the evaluation set on every slide, yet at $a_{\mathrm{spike}} = 2$ the residual policy remained below the feasible design, by 0.021 (0.012 to 0.030) PCC at 5% and 0.011 (0.004 to 0.019) at 10%, close to its deficits without injection (0.015 and 0.007; Figure 2c, Appendix H). Feedback still helped relative to permuted values, by $+0.001$ to $+0.013$. The protocol can thus register a large detection effect, but not demonstrably a reconstruction gain, and the RQ1 results below concern the tested policies.

Table 1: Reconstruction accuracy (PCC over the top-50 genes on unprofiled regions), mean over 22 slides. "Feasible" is the non-adaptive design chosen on the other studies; "hindsight" is the hindsight benchmark. "Residual, perm." is the residual policy given permuted feedback.

	Non-adaptive		Sequential fixed design		Feedback		
Budget	feasible	hindsight	error-red.	max-unc.	residual	residual, perm.	local-var.
0.5%	0.449	0.470	0.343	0.348	0.357	0.327	0.348
1%	0.534	0.550	0.467	0.462	0.477	0.438	0.424
2%	0.591	0.607	0.567	0.562	0.566	0.536	0.471
5%	0.659	0.660	0.650	0.648	0.643	0.633	0.513
10%	0.685	0.687	0.684	0.685	0.678	0.677	0.535
20%	0.707	0.709	0.708	0.710	0.705	0.705	0.544

Table 2: Reconstruction margins (PCC), 22 slides, 18 studies. Δ: mean per-slide difference with an unadjusted 95% CI resampling studies. p_H: Holm-adjusted within the stage-2 primary or secondary family; p: unadjusted (post hoc).

	residual − feasible (primary)		residual − residual, perm. (secondary)		error-red. − feasible (post hoc)	
Budget	Δ [95% CI]	p_H	Δ [95% CI]	p_H	Δ [95% CI]	p
0.5%	-0.092 $[-0.132, -0.054]$	0.002	$+0.031$ $[+0.010, +0.057]$	0.056	-0.107 $[-0.149, -0.067]$	<0.001
1%	-0.057 $[-0.089, -0.028]$	0.010	$+0.039$ $[+0.021, +0.060]$	0.002	-0.067 $[-0.093, -0.044]$	<0.001
2%	-0.025 $[-0.041, -0.009]$	0.099	$+0.030$ $[+0.014, +0.049]$	0.014	-0.024 $[-0.039, -0.007]$	0.013
5%	-0.016 $[-0.025, -0.009]$	0.003	$+0.009$ $[+0.001, +0.018]$	0.238	-0.010 $[-0.017, -0.003]$	0.021
10%	-0.007 $[-0.011, -0.003]$	0.010	$+0.001$ $[-0.002, +0.004]$	1.000	-0.001 $[-0.006, +0.003]$	0.763
20%	-0.002 $[-0.004, -0.001]$	0.126	-0.001 $[-0.003, +0.002]$	1.000	$+0.001$ $[-0.002, +0.003]$	0.537

4.2 RQ1: Reconstruction

No feedback policy outperformed the feasible design (Tables 1 and 2, Figure 2a). The residual policy fell below it at every budget, by 0.092 PCC at 0.5%, 0.057 at 1%, 0.025 at 2%, 0.016 at 5%, 0.007 at 10%, and 0.002 at 20%, significantly so at 0.5, 1, 5, and 10% ($p_H \leq 0.010$). The local-variance policy was further below, by 0.119 to 0.163 (secondary, $p_H \leq 0.001$). Bootstrap intervals that repeat the comparator's selection in each replicate excluded zero at every budget (post hoc; Appendix F).

The measured values nonetheless carried information. With informative rather than permuted feedback, the residual policy gained 0.031, 0.039, and 0.030 PCC at 0.5–2% (secondary, Appendix G; p_H 0.056, 0.002, 0.014), with no difference at 10–20%. Feedback moved the policy toward the feasible design without closing the gap. The error-reduction design, which shares the residual policy's structure but uses value-independent uncertainty, trailed the feasible design by similar amounts (0.107, 0.067, and 0.024 at 0.5–2%; post hoc), and both converged on it as the budget grew. The only design to exceed the feasible design was maximum-uncertainty, by 0.003 PCC at 20% (95% CI 0.002 to 0.004, $p = 0.0007$, post hoc), a fixed design that uses no feedback. The hindsight benchmark exceeded the feasible design by 0.016 to 0.021 PCC at budgets up to 2% and by at most 0.002 above (Appendix E).

4.3 RQ2: Detection

Informative feedback raised the local-variance policy's recall over permuted feedback by 0.005, 0.017, 0.029, and 0.063 at 2, 5, 10, and 20% (Table 3, Figure 2b). The unadjusted bootstrap intervals excluded zero at 5–20%, but intervals from inverting the sign-flip test did so only at 20% (Appendix M); no margin survived adjustment ($p_H \geq 0.29$), and each estimate lay below its MDE_H (0.015, 0.035, 0.066, 0.112). The margin over the feasible design was smaller (0.006 to 0.026, $p_H \geq 0.25$), and the margin over the diversity sampler was close to zero (0.001 to 0.012, $p_H = 1.000$).

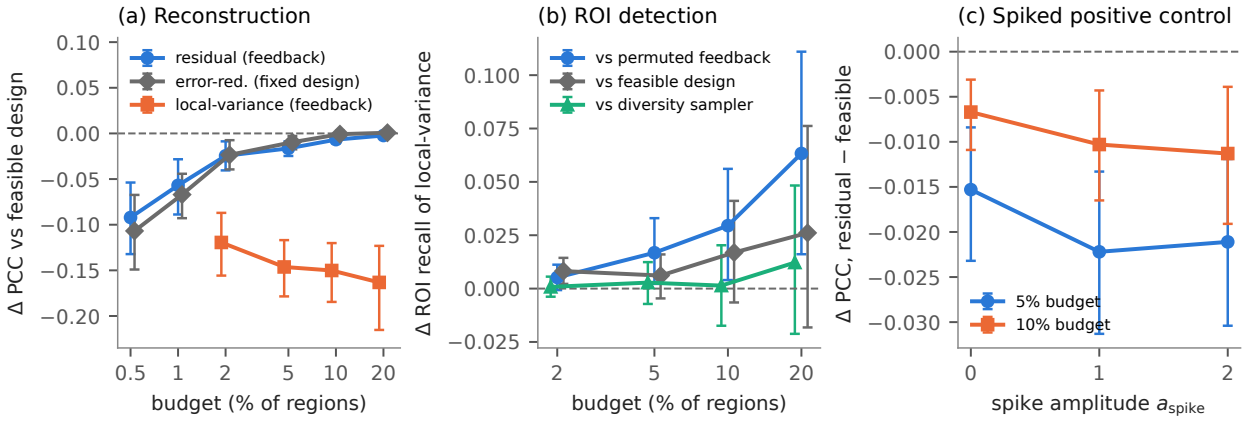


Figure 2: Stage-2 margins over 22 slides, mean and unadjusted 95% CI resampling studies. (a) Reconstruction margin over the feasible design. (b) Detection-recall margin of the local-variance policy over its permuted-feedback version, the feasible design, and the diversity sampler. (c) Spiked control: residual policy minus feasible design by spike amplitude.

Table 3: Detection margins (ROI recall) of the local-variance policy, 22 slides. Δ with unadjusted 95% CI resampling studies; p_H Holm-adjusted over the 18 stage-2 primary margins; MDE_H at 80% power and $\alpha = 0.05/18$.

	local-var. − permuted feedback			local-var. − feasible		local-var. − diversity	
Budget	Δ [95% CI]	p_H	MDE_H	Δ [95% CI]	p_H	Δ [95% CI]	p_H
2%	+0.005 [−0.001, +0.011]	1.000	0.015	+0.008 [+0.002, +0.014]	0.250	+0.001 [−0.004, +0.006]	1.000
5%	+0.017 [+0.002, +0.033]	0.530	0.035	+0.006 [−0.005, +0.016]	1.000	+0.003 [−0.007, +0.012]	1.000
10%	+0.029 [+0.004, +0.056]	0.530	0.066	+0.017 [−0.006, +0.041]	1.000	+0.001 [−0.017, +0.020]	1.000
20%	+0.063 [+0.016, +0.111]	0.294	0.112	+0.026 [−0.018, +0.076]	1.000	+0.012 [−0.021, +0.048]	1.000

All designs stayed close to random selection, which captures on average the budget fraction of the ROIs (Table 4). At 20% the local-variance policy exceeded this expectation by 0.078 recall (95% CI from sign-flip inversion 0.029 to 0.125) and the feasible design by 0.052, while at 2% the feasible design was slightly below it (0.018 against 0.020). The oracle reached 0.999 at 20%, so every policy captured only a small part of the heterogeneity available to it (post hoc; Appendix M).

The absolute recalls locate the feedback effect (Table 4). At 20%, the policy recovered 0.278 of the ROIs with informative feedback but only 0.215 with permuted and 0.204 with constant values, below the feasible design (0.252); the diversity sampler reached 0.266 without feedback. Part of the feedback effect therefore recovers ground that the selection rule loses on its own. The margin over constant feedback at 20% was the only significant detection effect of feedback (+0.074, $p_H = 0.038$, secondary).

The apparent advantage also depends on the comparator. Against MI-greedy used on every slide, the local-variance policy gained 0.043 and 0.078 recall at 10 and 20% ($p_H = 0.043$ for both; post hoc). MI-greedy, however, is a reconstruction design whose detection recall never exceeded that expected of random selection (0.200 at 20%, against 0.247 for variance-greedy and 0.252 for the one-shot ranker), so this comparison shows only that feedback beats chance-level placement. Selection-aware intervals against the feasible design included zero at every budget (Appendix F). The ROI-seeking port did not improve on the coverage designs in stage 1 (Appendix D). Because the port is not the trained SOFisher, we also ran SOFisher natively on its mouse cortex data, where its reward for captured target regions rose from a median of −238 early in training to +147 late (descriptive; Appendix L); the port's result is therefore not a test of SOFisher. SCR^2-ST could not be run natively because its processed training data were not public when we ran our experiments.

Table 4: ROI-detection recall, mean over 22 slides. "Random" is the recall expected of random selection (n/N for n of N regions). "Informative", "permuted", and "constant" denote the local-variance policy with informative, permuted, and constant feedback. The oracle profiles regions in order of true heterogeneity.

		Non-adaptive				Feedback (local variance)			
Budget	random	feasible	k-means	MI-greedy	diversity	informative	permuted	constant	oracle
2%	0.020	0.018	0.018	0.018	0.026	0.027	0.021	0.022	0.100
5%	0.050	0.064	0.050	0.046	0.067	0.070	0.053	0.051	0.250
10%	0.100	0.120	0.110	0.094	0.136	0.137	0.107	0.103	0.500
20%	0.200	0.252	0.232	0.200	0.266	0.278	0.215	0.204	0.999

Table 5: Stage-3 primary margins: hybrid minus its base design run for the whole budget, 22 slides. The 50% and 80% hybrids take that share of regions from the base design and the rest from the feedback policy. Δ with unadjusted 95% CI resampling studies; p_H Holm-adjusted over the 20 primary margins; MDE_H at 80% power and $\alpha = 0.05/20$.

		50% hybrid			80% hybrid		
Budget	base design	Δ [95% CI]	p_H	MDE_H	Δ [95% CI]	p_H	MDE_H
Reconstruction (PCC); feedback: residual policy							
0.5%	k-means	-0.036 $[-0.057, -0.018]$	0.006	0.056	-0.009 $[-0.021, +0.003]$	1.000	0.033
1%	k-means	-0.010 $[-0.020, +0.002]$	1.000	0.021	-0.009 $[-0.019, -0.001]$	1.000	0.025
2%	MI-greedy	$+0.001$ $[-0.007, +0.011]$	1.000	0.017	$+0.007$ $[+0.002, +0.013]$	0.359	0.013
5%	MI-greedy	-0.003 $[-0.007, +0.002]$	1.000	0.013	-0.001 $[-0.004, +0.003]$	1.000	0.008
10%	MI-greedy	0.000 $[-0.002, +0.002]$	1.000	0.005	$+0.001$ $[-0.001, +0.002]$	1.000	0.004
20%	MI-greedy	-0.002 $[-0.004, +0.000]$	1.000	0.005	-0.001 $[-0.003, +0.001]$	1.000	0.006
Detection (ROI recall); feedback: local-variance policy							
2%	variance-greedy	$+0.003$ $[+0.000, +0.005]$	0.756	0.007	0.000 $[-0.001, +0.002]$	1.000	0.004
5%	variance-greedy	$+0.002$ $[-0.007, +0.010]$	1.000	0.020	0.000 $[-0.003, +0.003]$	1.000	0.008
10%	variance-greedy	$+0.006$ $[-0.005, +0.018]$	1.000	0.026	$+0.001$ $[-0.005, +0.007]$	1.000	0.015
20%	one-shot	0.000 $[-0.033, +0.034]$	1.000	0.068	-0.003 $[-0.019, +0.014]$	1.000	0.035

4.4 RQ3: Feedback on top of a fixed design

No hybrid outperformed its base design (Table 5, Figure 3). For reconstruction, the hybrids recovered most of the residual policy's deficit, but the 50% hybrid remained below k-means at 0.5%, by 0.036 PCC ($p_H = 0.006$), and from 2% upward every hybrid was within 0.007 PCC of MI-greedy with no significant margin. For detection, every hybrid margin lay between -0.003 and $+0.006$ recall ($p_H \geq 0.76$). These nulls are tight: at 2–10% the MDE_H values were 0.004 to 0.017 PCC and 0.004 to 0.026 recall, so gains above about 0.02 PCC or 0.03 recall would probably have been detected. At 20% the detection bound widens to 0.035–0.068.

The hybrids responded to the values they were shown. With permuted values they did worse at 2%, by 0.016 and 0.007 PCC for the two reconstruction hybrids and 0.007 recall for the 50% detection hybrid (secondary, $p_H \leq 0.027$). For that detection hybrid the difference grew to 0.014–0.035 at 5–20% without surviving correction; for reconstruction at 10–20% the permuted versions were marginally ahead, by at most 0.003 PCC ($p_H \geq 0.12$; Appendix I). In absolute terms (Table 16), with permuted values the feedback phase lost ground relative to continuing the base design, for example from 0.127 to 0.110 recall at 10%, and informative values brought the hybrid back to about the level of the design (0.133). Because permuted values can actively mislead, this shows that informative values did better than wrong ones, not that they added to the design. In detection, the local-variance policy run alone (0.137 at 10%, 0.278 at 20%) scored above both hybrids at every budget (descriptive).

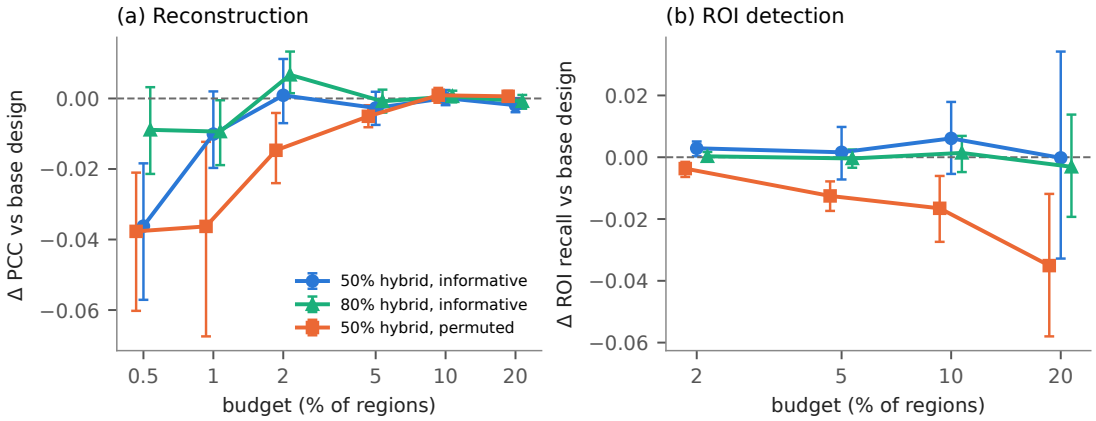


Figure 3: Hybrid minus its base design, 22 slides, mean and unadjusted 95% CI resampling studies. "Informative" hybrids receive the measured values (primary); the permuted series is the 50% hybrid given permuted feedback, averaged over two seeds (descriptive). Base designs as in Table 5.

Table 6: Batched acquisition, stage 3: local-variance policy revealing regions in rounds of 5, 22 slides. "One at a time" is the stage-2 sequential run. Δ with unadjusted 95% CI resampling studies; p_H Holm-adjusted over the 24 stage-3 secondary margins; permuted values averaged over two seeds.

		ROI recall				rounds of 5 − base design		rounds of 5 − permuted	
Budget	base design	base	one at a time	rounds of 5	rounds of 5, perm.	Δ [95% CI]	p_H	Δ [95% CI]	p_H
10%	variance-greedy	0.127	0.137	0.150	0.100	+0.023 [−0.000, +0.046]	0.749	+0.050 [+0.022, +0.075]	0.113
20%	one-shot	0.252	0.278	0.288	0.204	+0.035 [−0.017, +0.095]	1.000	+0.084 [+0.025, +0.141]	0.240

4.5 RQ4: Batched and contiguous acquisition (exploratory)

In stage 1, at 10% budget, the local-variance policy's detection margins over the hindsight benchmark of the coverage designs were +0.021, +0.034, and +0.011 for rounds of 1, 5, and 25 regions and +0.024 for contiguous blocks, none with $p_H < 0.05$ (Appendix D). With a single round the policy has no measurements and selects the regions nearest the slide centroid, its rule while no region is measured; the resulting margin of −0.051 ($p_H = 0.008$) reflects that rule rather than adaptivity. Reconstruction margins were negative in every setting.

In stage 3, rounds of 5 regions did not reduce recall: the policy captured 0.150 and 0.288 of ROIs at 10 and 20%, slightly more than when choosing one region at a time (0.137 and 0.278; Table 6). Its margins over the base detection design (+0.023, +0.035) and over permuted feedback (+0.050, +0.084) did not survive correction ($p_H \geq 0.11$).

4.6 Robustness

Permutation seeds. Repeating the stage-2 ablations with further seeds (descriptive; Appendix I) did not change any conclusion. Across seeds the mean margin varied with a standard deviation of at most 0.007 PCC and 0.013 recall. The seed-averaged detection margins (+0.005, +0.017, +0.033, +0.074 at 2–20%) stayed below their MDE_H values, and for reconstruction the registered seed gave the largest of three margins at 0.5% (+0.031, against +0.024 and +0.016).

Evaluation genes. Choosing the top-50 genes from the whole slide uses information a prospective study would lack. Choosing them from a random 15% reference subset changed the stage-1 reconstruction margins by at most 0.003, and scoring all filtered genes left them negative (−0.033 to −0.079; Appendix J).

Platform. In an exploratory split, stage-1 detection margins were larger on Visium HD than on Visium (+0.048 against +0.008 at 10%), but with five HD studies the smallest attainable p is 0.0625, and platform is confounded with study (Appendix K).

5 Discussion

Reconstruction. Informative feedback beat permuted feedback by up to 0.039 PCC when it drove the whole selection and by up to 0.026 when it chose only the remainder, yet neither arrangement improved on a fixed design. Permuted values can mislead a policy as well as fail to inform it, so these contrasts show that the policies responded to the values, not how much the values were worth; the cleaner constant-feedback control was run only for the detection policy. The pattern fits sensor-placement theory, under which placement matters more than the observed values when uncertainty depends on locations alone (Krause et al., 2008). Our imputer's uncertainty has this property while its predictions do not, so any benefit of feedback had to come through the predictions. The evaluation does not test the theory, and because the semi-synthetic control failed it cannot rule out a better feedback policy. What it establishes is that the residual policy, which targets the imputer's own error and received exact measurements, lost to a fixed design chosen without the target slide.

Detection. Feedback beat weak baselines but not strong ones. It outperformed MI-greedy, a reconstruction design that performed at chance level for detection (post hoc), and, at the largest budget, its own constant-feedback version. Such gains are what active-search reasoning predicts (Garnett et al., 2012), and the synthetic control confirmed that the simulator detects them. Against a design chosen for detection or a diversity sampler, the margins were small and not significant, and spending part of the budget on a fixed design before feedback did not help. The diversity sampler's near-parity indicates that, on these slides, heterogeneous regions are spread widely enough that broad coverage finds many of them; every design, adaptive or not, stayed close to random selection. Resolving a difference of the observed size would need more independent studies than the 18 used here, especially at 20%.

Implications. For developers of adaptive acquisition, the lesson is methodological: beating random selection is not evidence that feedback helps. Two comparisons should be standard: a fixed design chosen on other data for the same objective, and the method itself fed uninformative values. Neither requires new data, and together they separate the value of feedback from the value of a good selection rule; inference should treat studies, not slides, as the unit. For experimentalists, within simulated spot-level acquisition at 0.5–20% of regions, one imputer, and computationally defined endpoints, no tested feedback policy outperformed a fixed design, so a fixed design is the practical default: k-means at the smallest budgets and MI-greedy above them for reconstruction, and variance-greedy, the one-shot ranker, or the diversity sampler for heterogeneity. These designs are computed before any measurement and avoid the turnaround that sequential selection requires. The negative result was also obtained under conditions generous to feedback, since simulated measurements are exact, arrive at once, and cost no turnaround. Whether it holds for ROI-scale acquisition, other predictors, or annotated biological targets would need prospective or annotated data. Our findings agree with the benchmark's result for predictor training (Zhu et al., 2026b), although the two studies address different decisions.

Limitations. *Simulation.* The simulator is retrospective: a revealed region returns its measured expression without added noise, batch effects, or delay, conditions that if anything favour feedback. The acquisition unit is a Visium spot or a 128 μm HD pseudo-spot, which no platform selects individually, and is smaller than a typical GeoMx ROI. The batch and block simulations approximate but do not validate any platform, and rounds of 5 regions at 10–20% budgets leave the policy close to fully sequential.

Endpoints. Reconstruction is scored on the regions each policy left unprofiled, so policies are compared on different region sets; because correlation falls when the evaluated values vary less, a policy that profiles the most variable regions is scored on a harder set for this metric, which could contribute to the deficits of the feedback policies. The ROIs are the 20% of regions with the highest local-variance score, not annotated structures; high local variance can mark tissue boundaries, mixtures of cell types, or technical noise as well

as targets such as tumour margins or immune niches, and all designs captured them at rates close to random selection. The detection results therefore concern computational heterogeneity and do not show whether feedback helps find a particular biological target.

Models and policies. The 22-slide results use one imputer, whose hyperparameters were set on one of the 22 slides (excluding it changed no conclusion; Appendix M); the k-NN and kriging comparisons rest on five slides. The two feedback policies are hand-built heuristics with settings chosen in advance rather than tuned, so the results bound what these policies achieve, not what feedback can achieve. The competitor samplers reimplement objectives; SOFisher was run natively only on its own data and SCR^2-ST not at all.

Inference. The registered stage-2 ablation tests use one permutation, with further seeds descriptive. Multiplicity is controlled within each registered family, not across the three stages or the post hoc analyses. The analysis plans were registered in our own version-controlled repository rather than with an external registry, so their timing rests on commit records, which the supplementary material lists; the feasible design, the residual policy, the ablations, and the hybrids were added after earlier results were known, although each was registered before its experiments. Budgets above 20% were not tested.

6 Conclusion

Across 22 human slides from 18 studies and three registered stages, the feedback policies we tested did not earn their measurement turnaround. For whole-section reconstruction, the residual policy was significantly worse than a well-chosen fixed design, and spending only the remainder of the budget on feedback did not improve on that design. For capturing heterogeneous regions, feedback helped relative to a design built for the other objective, but not detectably relative to a design built for detection or to a diversity sampler. Feasible objective-matched baselines, uninformative-feedback controls, and study-level inference with positive controls and minimum detectable effects together separate the value of feedback from the value of a good selection rule, and new adaptive acquisition methods should be judged against them.

Reproducibility. The supplementary material contains the code, configuration, result files, the registered analysis plans with their commit dates, and the scripts that regenerate every table and figure from the result files. A public repository will be linked on acceptance. HEST-1k and UNI2-h are available from their providers under their own access terms.

References

Douglas G. Altman and J. Martin Bland. Absence of evidence is not evidence of absence. *BMJ*, 311(7003): 485, 1995. doi: 10.1136/bmj.311.7003.485.

Benjamin F. Arnold, Daniel R. Hogan, John M. Colford, and Alan E. Hubbard. Simulation methods to estimate design power: an overview for applied research. *BMC Medical Research Methodology*, 11:94, 2011. doi: 10.1186/1471-2288-11-94.

Christoph Bernau, Markus Riester, Anne-Laure Boulesteix, Giovanni Parmigiani, Curtis Huttenhower, Levi Waldron, and Lorenzo Trippa. Cross-study validation for the assessment of prediction algorithms. *Bioinformatics*, 30(12):i105–i112, 2014. doi: 10.1093/bioinformatics/btu279.

Howard S. Bloom. Minimum detectable effects: A simple way to report the statistical power of experimental designs. *Evaluation Review*, 19(5):547–556, 1995. doi: 10.1177/0193841X9501900504.

A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427, 2008. doi: 10.1162/rest.90.3.414.

Gavin C. Cawley and Nicola L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107, 2010.

Richard J. Chen, Tong Ding, Ming Y. Lu, Drew F. K. Williamson, Guillaume Jaume, Andrew H. Song, Bowen Chen, Andrew Zhang, Daniel Shao, Muhammad Shaban, et al. Towards a general-purpose foundation model for computational pathology. *Nature Medicine*, 30(3):850–862, 2024. doi: 10.1038/s41591-024-02857-3.

Jacob Cohen. *Statistical Power Analysis for the Behavioral Sciences*. Lawrence Erlbaum Associates, Hillsdale, NJ, 2nd edition, 1988.

Ian Covert, Rohan Gala, Tim Wang, Karel Svoboda, Uygar Sümbül, and Su-In Lee. Predictive and robust gene selection for spatial transcriptomics. *Nature Communications*, 14:2091, 2023. doi: 10.1038/s41467-023-37392-1.

Michelli Faria de Oliveira, Juan Pablo Romero, Meii Chung, Stephen R. Williams, Andrew D. Gottscho, et al. High-definition spatial transcriptomic profiling of immune cell populations in colorectal cancer. *Nature Genetics*, 57(6):1512–1523, 2025. doi: 10.1038/s41588-025-02193-3.

Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, New York, 1993.

Christopher A. Field and Alan H. Welsh. Bootstrapping clustered data. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 69(3):369–390, 2007. doi: 10.1111/j.1467-9868.2007.00593.x.

Roman Garnett, Yamuna Krishnamurthy, Xuehan Xiong, Jeff Schneider, and Richard Mann. Bayesian optimal active search and surveying. In *Proceedings of the 29th International Conference on Machine Learning (ICML)*, 2012.

Damien Garreau, Wittawat Jitkrittum, and Motonobu Kanagawa. Large sample analysis of the median heuristic. *arXiv preprint arXiv:1707.07269*, 2017.

Daniel Golovin and Andreas Krause. Adaptive submodularity: Theory and applications in active learning and stochastic optimization. *Journal of Artificial Intelligence Research*, 42:427–486, 2011.

Teofilo F. Gonzalez. Clustering to minimize the maximum intercluster distance. *Theoretical Computer Science*, 38:293–306, 1985. doi: 10.1016/0304-3975(85)90224-5.

Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, New York, 2nd edition, 2009.

Tomislav Hengl, Gerard B. M. Heuvelink, and David G. Rossiter. About regression-kriging: From equations to case studies. *Computers & Geosciences*, 33(10):1301–1315, 2007. doi: 10.1016/j.cageo.2007.05.001.

Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6 (2):65–70, 1979.

Tinglin Huang, Tianyu Liu, Mehrtash Babadi, Wengong Jin, and Rex Ying. Scalable generation of spatial transcriptomics from histology images via whole-slide flow matching. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025.

Guillaume Jaume, Paul Doucet, Andrew H. Song, Ming Y. Lu, Cristina Almagro-Pérez, Sophia J. Wagner, Anurag J. Vaidya, Richard J. Chen, Drew F. K. Williamson, Ahrong Kim, and Faisal Mahmood. HEST-1k: A dataset for spatial transcriptomics and histology image analysis. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024.

Donald R. Jones, Matthias Schonlau, and William J. Welch. Efficient global optimization of expensive black-box functions. *Journal of Global Optimization*, 13(4):455–492, 1998. doi: 10.1023/A:1008306431147.

Andreas Kirsch, Joost van Amersfoort, and Yarin Gal. BatchBALD: Efficient and diverse batch acquisition for deep Bayesian active learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.

Andreas Krause, Ajit Singh, and Carlos Guestrin. Near-optimal sensor placements in Gaussian processes: Theory, efficient algorithms and empirical studies. *Journal of Machine Learning Research*, 9:235–284, 2008.

David D. Lewis and William A. Gale. A sequential algorithm for training text classifiers. In *Proceedings of the 17th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 3–12, 1994. doi: 10.1007/978-1-4471-2099-5_1.

Zhuo Li, Weiran Wu, Chuangyi Han, Yan Cui, Tian Lu, Rongqin Ke, Jian Sun, and Zhiyuan Yuan. SOFisher: reinforcement learning-guided experiment designs for spatial omics. *Nature Communications*, 17:6830, 2026. doi: 10.1038/s41467-026-73404-6.

Stuart P. Lloyd. Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 28(2): 129–137, 1982. doi: 10.1109/TIT.1982.1056489.

Haoran Lu, Luyang Fang, Orlando Zeng, Wenxuan Zhong, Guo-Cheng Yuan, and Ping Ma. Optimal gene panel selection for targeted spatial transcriptomics experiments. *Nucleic Acids Research*, 54(11):gkag621, 2026. doi: 10.1093/nar/gkag621.

Christopher R. Merritt, Giang T. Ong, Sarah E. Church, Kristi Barker, Patrick Danaher, Gary Geiss, Margaret Hoang, Jaemyeong Jung, Yan Liang, Jill McKay-Fleisch, et al. Multiplex digital spatial profiling of proteins and RNA in fixed tissue. *Nature Biotechnology*, 38(5):586–599, 2020. doi: 10.1038/s41587-020-0472-9.

Lambda Moses and Lior Pachter. Museum of spatial transcriptomics. *Nature Methods*, 19(5):534–546, 2022. doi: 10.1038/s41592-022-01409-2.

Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018. doi: 10.1073/pnas.1708274114.

Markus Ojala and Gemma C. Garriga. Permutation tests for studying classifier performance. *Journal of Machine Learning Research*, 11:1833–1863, 2010.

Jiarui Ouyang, Yihui Wang, Yihang Gao, Yingxue Xu, Shu Yang, and Hao Chen. GenAR: Next-scale autoregressive generation for spatial gene expression prediction. *arXiv preprint arXiv:2510.04315*, 2025.

Ali Rahimi and Benjamin Recht. Random features for large-scale kernel machines. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 20, 2007.

Anjali Rao, Dalia Barkley, Gustavo S. França, and Itai Yanai. Exploring tissue architecture using spatial transcriptomics. *Nature*, 596(7871):211–220, 2021. doi: 10.1038/s41586-021-03634-9.

Nicholas Roy and Andrew McCallum. Toward optimal active learning through sampling estimation of error reduction. In *Proceedings of the 18th International Conference on Machine Learning (ICML)*, pp. 441–448, 2001.

Ozan Sener and Silvio Savarese. Active learning for convolutional neural networks: A core-set approach. In *International Conference on Learning Representations (ICLR)*, 2018.

Burr Settles. Active learning literature survey. Computer Sciences Technical Report 1648, University of Wisconsin–Madison, 2009.

Patrik L. Ståhl, Fredrik Salmén, Sanja Vickovic, Anna Lundmark, José Fernández Navarro, et al. Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. *Science*, 353(6294): 78–82, 2016. doi: 10.1126/science.aaf2403.

Valentine Svensson, Sarah A. Teichmann, and Oliver Stegle. SpatialDE: identification of spatially variable genes. *Nature Methods*, 15(5):343–346, 2018. doi: 10.1038/nmeth.4636.

Sudhir Varma and Richard Simon. Bias in error estimation when using cross-validation for model selection. *BMC Bioinformatics*, 7:91, 2006. doi: 10.1186/1471-2105-7-91.

Anderson M. Winkler, Gerard R. Ridgway, Matthew A. Webster, Stephen M. Smith, and Thomas E. Nichols. Permutation inference for the general linear model. *NeuroImage*, 92:381–397, 2014. doi: 10.1016/j.neuroimage.2014.01.060.

Ronald Xie, Kuan Pang, Sai W. Chung, Catia T. Perciani, Sonya A. MacParland, Bo Wang, and Gary D. Bader. Spatially resolved gene expression prediction from histology images via bi-modal contrastive learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

Jonathan Xu, Michelle Jiang, Shunsuke Koga, Nancy Zhang, and Zhi Huang. SpatialFinder: a human-in-the-loop vision-language framework for prioritizing high-value regions in spatial transcriptomics. *Frontiers in Bioinformatics*, 6:1746714, 2026. doi: 10.3389/fbinf.2026.1746714.

Haiyan Yin, Yingzhen Li, Sinno Jialin Pan, Cheng Zhang, and Sebastian Tschiatschek. Reinforcement learning with efficient active feature acquisition. *arXiv preprint arXiv:2011.00825*, 2020.

Musu Yuan, Kaitian Jin, Hanying Yan, Amelia Schroeder, Chunyu Luo, Sicong Yao, Bernhard Dumoulin, Jonathan Levinsohn, Tianhao Luo, Jean R. Clemenceau, et al. Smart spatial omics (S2-omics) optimizes region of interest selection to capture molecular heterogeneity in diverse tissues. *Nature Cell Biology*, 27: 2225–2238, 2025. doi: 10.1038/s41556-025-01811-w.

Junchao Zhu, Ruining Deng, Junlin Guo, Tianyuan Yao, Chongyu Qu, Juming Xiong, Siqi Lu, Zhengyi Lu, Yanfan Zhu, Marilyn Lionts, Yuechen Yang, Yalin Zheng, Yu Wang, Shilin Zhao, Haichun Yang, and Yuankai Huo. SCR^2-ST: Combine single cell with spatial transcriptomics for efficient active sampling via reinforcement learning. In *Proceedings of the 9th International Conference on Medical Imaging with Deep Learning (MIDL)*, volume 315 of *Proceedings of Machine Learning Research*, pp. 488–509. PMLR, 2026a.

Sichen Zhu, Yuchen Zhu, Molei Tao, and Peng Qiu. Diffusion generative modeling for spatially resolved gene expression inference from histology images. In *International Conference on Learning Representations (ICLR)*, 2025.

Xiaojin Zhu, Zoubin Ghahramani, and John D. Lafferty. Semi-supervised learning using Gaussian fields and harmonic functions. In *Proceedings of the 20th International Conference on Machine Learning (ICML)*, pp. 912–919, 2003.

Zheyu Zhu, Junchao Zhu, Fengbei Liu, Tianyuan Yao, Gelei Xu, John Cannon, Haichun Yang, Yuankai Huo, Mert R. Sabuncu, and Ruining Deng. Benchmarking active spot selection for cost-efficient spatial transcriptomics. *arXiv preprint arXiv:2609.27208*, 2026b.

A Slides

Table 7 lists the slides. "Cluster" indexes the HEST study used as the resampling unit, and "Pitch" is the median nearest-neighbour spacing of regions.

Table 7: The 22 slides (HEST-1k v1.3 metadata).

Tissue	HEST ID	Platform	Spots	μm/px	Pitch (μm)	Cluster	Study (HEST dataset title)
prostate	TENX174	Visium HD	2385	0.274	128	1	Visium HD Spatial Gene Expression Library, Human Prostate ...
prostate	TENX40	Visium	4371	0.345	100	2	Human Prostate Cancer, Adenocarcinoma with Invasive Carcin...
prostate	MEND160	Visium	4079	0.573	100	3	Spatially resolved clonal copy number alterations in benig...
prostate	TENX46	Visium	3043	0.344	100	4	Human Prostate Cancer, Acinar Cell Carcinoma (FFPE)
prostate	TENX41	Visium	2543	0.345	100	5	Normal Human Prostate (FFPE)
prostate	MEND144	Visium	4074	0.341	100	3	Spatially resolved clonal copy number alterations in benig...
kidney	TENX173	Visium HD	2684	0.274	128	6	Visium HD Spatial Gene Expression Library, Human Kidney (F...
kidney	TENX71	Visium	5936	0.274	100	7	Human Kidney, 11 mm Capture Area (FFPE)
kidney	INT21	Visium	4975	0.457	100	8	Tertiary lymphoid structures generate and propagate anti-t...
kidney	INT19	Visium	4948	0.457	100	8	Tertiary lymphoid structures generate and propagate anti-t...
lung	TENX171	Visium HD	1766	0.274	128	9	Visium HD Spatial Gene Expression Libraries, Post-Xenium, ...
lung	TENX170	Visium HD	1877	0.274	128	9	Visium HD Spatial Gene Expression Libraries, Post-Xenium, ...
lung	TENX163	Visium HD	2487	0.274	128	10	Visium HD Spatial Gene Expression Library, Human Lung Canc...
lung	TENX72	Visium	6195	0.274	100	11	Human Lung Cancer, 11 mm Capture Area (FFPE)
lung	TENX62	Visium	3858	0.273	100	12	Human Lung Cancer (FFPE)
lung	MISC20	Visium	3316	0.438	100	13	A spatially resolved atlas of the human lung characterizes...
colon	TENX156	Visium HD	2179	0.274	128	14	Characterization of immune cell populations in the tumor m...
colon	TENX155	Visium HD	2281	0.274	128	14	Characterization of immune cell populations in the tumor m...
colon	TENX70	Visium	9080	0.274	100	15	Human Colorectal Cancer, 11 mm Capture Area (FFPE)
colon	TENX92	Visium	6518	0.274	100	16	Visium CytAssist Gene Expression Libraries of Post-Xenium ...
colon	MISC47	Visium	4818	0.276	100	17	COLON MAP: Colon Molecular Atlas Project
colon	ZEN48	Visium	2385	0.459	100	18	Charting the Heterogeneity of Colorectal Cancer Consensus ...

Changes from an earlier seven-slide analysis. An earlier analysis used seven Visium HD slides and described regions as 112 μm squares. The current HEST metadata lists two of those slides as non-human (TENX186, *Mus musculus*; TENX175, mixed human and mouse), and both are excluded here. The loader had also assumed 0.5 μm/px, whereas the HD images are about 0.274 μm/px, so those regions were about 61 μm squares containing one pseudo-spot each. On the five retained slides, per-slide local-variance margins (reconstruction and detection, four budgets) differ between the two analyses by a median of 0.015 (maximum 0.096); 3 of 40 change sign, two by less than 0.005 and one detection margin from -0.050 to $+0.015$. With seven slides the smallest attainable exact p-value is 0.016, so no earlier margin could have survived correction for 24 comparisons.

B Configuration and implementation checks

All values are fixed in the released `config.py`. The imputer's hyperparameters were set during development on one prostate slide (TENX174, one of the 22) before any registered analysis and were not changed afterwards; the policies' hyperparameters were fixed before the stage in which each policy first appears.

Seeds: global 2021; random policy $\{0, \ldots, 4\}$; the stage-2 permutation uses the global seed, and stage 3 uses seeds 1 and 2 (1–4 for the repeated local-variance ablation). Imputer: image PCA 64, random Fourier features 512 (median-heuristic bandwidth), ridge 10, graph $k = 8$, 20 propagation steps, step size 0.5. k-NN imputer: $k = 10$, joint image and space features. Kriging: image PCA 64, ridge 10, residual $k = 10$. Placement kernels: feature PCA 64, median-heuristic lengthscale, noise 10^{-2}. Error-reduction and residual policies: neighbourhood 15; the residual policy uses the 256 genes with the highest variance among profiled regions and 10 profiled neighbours. Local-variance: spatial kernel width 0.1 of the slide diagonal, exploration weight 0.5. Hybrids: coverage fraction $f \in \{0.5, 0.8\}$, rounded up. Batched stage-3 runs: rounds of 5. Spiked control: 20 genes, 3 hotspots, radius 0.05 of the diagonal, noise 0.1 of the reference standard deviation. Statistics: 5,000 bootstrap resamples; exact sign-flip enumeration over studies; MDE from 300 study resamples with 3,000 sign flips each, on a grid of 51 effect sizes.

The code for the larger slides replaces dense distance computations with sparse and incremental ones; a 132-run check (11 policies, 3 imputers, 2 slide sizes, 2 budgets) reproduced the original code's selections and scores exactly. In the stage-2 run, the five non-adaptive designs reproduced the stage-1 per-slide scores exactly, and the local-variance and diversity policies reproduced their stage-1 means. A caching error interrupted the first stage-2 run. For the results saved before it, a stale cache entry could only have produced a crash, and the error was fixed before any result was analysed.

C Imputer comparison on the earlier slide set

The k-NN and kriging imputers were run only on the earlier slide set. Table 8 reports error-reduction minus the hindsight benchmark on the five human slides from that run, whose regions held one pseudo-spot each, with the number of slides on which the margin was positive. With five slides these values are descriptive.

Table 8: Error-reduction minus the hindsight benchmark (PCC), mean over 5 human Visium HD slides (positive/total).

Budget	kNN	kriging	graphflow
0.5%	−0.088 (0/5)	−0.088 (2/5)	−0.082 (0/5)
1%	−0.114 (0/5)	−0.138 (1/5)	−0.083 (0/5)
2%	−0.091 (0/5)	−0.150 (1/5)	−0.040 (0/5)
5%	−0.121 (0/5)	−0.141 (1/5)	−0.011 (1/5)
10%	−0.134 (0/5)	−0.133 (0/5)	−0.000 (2/5)
20%	−0.145 (0/5)	−0.126 (0/5)	−0.002 (2/5)

D Stage-1 results

Stage 1 compared policies with hindsight benchmarks. Table 9 gives its 24-margin primary family except the batched and contiguous settings, which are in Table 10; Table 11 gives absolute recall, including the ROI-seeking sampler.

Table 9: Stage-1 primary margins against hindsight benchmarks: for error-reduction, the best of the five non-adaptive designs on each slide; for local-variance, the best of the three coverage designs. Unadjusted 95% CIs resampling studies; p_H Holm-adjusted over the 24 stage-1 primary margins.

	Reconstruction (PCC, top-50 genes)				Detection (ROI recall)	
	error-reduction − hindsight (non-adapt.)		local-var. − hindsight (coverage)		local-var. − hindsight (coverage)	
Budget	Δ [95% CI]	p_H	Δ [95% CI]	p_H	Δ [95% CI]	p_H
0.5%	−0.128 [−0.162, −0.096]	<0.001	–	–	–	–
1%	−0.083 [−0.105, −0.064]	<0.001	–	–	–	–
2%	−0.040 [−0.054, −0.027]	<0.001	−0.136 [−0.171, −0.106]	<0.001	+0.001 [−0.004, +0.006]	0.723
5%	−0.011 [−0.018, −0.004]	0.057	−0.148 [−0.180, −0.119]	<0.001	+0.013 [+0.002, +0.024]	0.362
10%	−0.002 [−0.007, +0.001]	0.687	−0.152 [−0.186, −0.122]	<0.001	+0.021 [+0.003, +0.037]	0.362
20%	−0.002 [−0.004, +0.001]	0.687	−0.165 [−0.217, −0.125]	<0.001	+0.042 [+0.006, +0.077]	0.362

Table 10: Stage-1 local-variance policy minus the hindsight benchmark of the coverage designs at 10% budget under batched and contiguous acquisition. Δ with unadjusted 95% CI resampling studies; p_H Holm-adjusted over the 24 stage-1 primary margins.

	Reconstruction (PCC)		Detection (ROI recall)	
Acquisition setting	Δ [95% CI]	p_H	Δ [95% CI]	p_H
Sequential (1 region per round)	-0.152 $[-0.186, -0.122]$	<0.001	$+0.021$ $[+0.003, +0.037]$	0.362
Rounds of 5 regions	-0.129 $[-0.175, -0.094]$	<0.001	$+0.034$ $[+0.013, +0.053]$	0.073
Rounds of 25 regions	-0.130 $[-0.176, -0.095]$	<0.001	$+0.011$ $[-0.000, +0.021]$	0.362
Single round (whole budget)	-0.236 $[-0.288, -0.188]$	<0.001	-0.051 $[-0.077, -0.027]$	0.008
Contiguous blocks of 7 regions	-0.087 $[-0.116, -0.063]$	<0.001	$+0.024$ $[+0.002, +0.044]$	0.362

Table 11: Stage-1 ROI-detection recall, mean over 22 slides. †Reimplementations of the stated objectives of SCR^2-ST (diversity, non-adaptive) and SOFisher (ROI-seeking, adaptive). *Ground-truth ordering.

	Non-adaptive coverage			Competitor objectives			
Budget	k-means	grid	MI-greedy	diversity†	ROI-seeking†	local-var.	oracle*
2%	0.018	0.022	0.018	0.026	0.021	0.027	0.100
5%	0.050	0.051	0.046	0.067	0.050	0.070	0.250
10%	0.110	0.103	0.094	0.136	0.104	0.137	0.500
20%	0.232	0.204	0.200	0.266	0.232	0.278	0.999

E Feasible design choices and the hindsight gap

The leave-one-study-out choice selected k-means for reconstruction at 0.5–1% and MI-greedy at 2–20% (with k-means for 4 of 18 studies at 2%), and for detection variance-greedy for most studies at 2–10% and the one-shot ranker at 20% (Table 12).

Table 12: Designs chosen on the other studies (number of held-out studies, of 18) and the difference between the hindsight benchmark and the feasible design for reconstruction (unadjusted 95% CI resampling studies).

Budget	feasible choice, reconstruction	feasible choice, detection	hindsight benchmark − feasible (PCC)
0.5%	kmeans (18)	–	$+0.021$ $[+0.006, +0.038]$
1%	kmeans (18)	–	$+0.016$ $[+0.006, +0.027]$
2%	mi-greedy (14), kmeans (4)	var-greedy (12), oneshot (5), grid (1)	$+0.017$ $[+0.008, +0.027]$
5%	mi-greedy (18)	var-greedy (18)	$+0.001$ $[+0.000, +0.003]$
10%	mi-greedy (18)	var-greedy (16), oneshot (2)	$+0.001$ $[+0.001, +0.003]$
20%	mi-greedy (18)	oneshot (18)	$+0.002$ $[+0.001, +0.003]$

F Inference for the feasible design

The feasible design for each held-out study is chosen from the other 17, so the 18 study margins share the selection step, which the study-level tests in the main text do not account for. Two post hoc checks address this (Table 13). The selection-aware bootstrap resamples studies with replacement and repeats the leave-one-study-out choice inside each replicate, excluding copies of the held-out study from its own selection, so its interval reflects uncertainty in which design is chosen. The single-design analysis uses MI-greedy on every slide in place of the feasible design, for which the sign-flip test is valid as specified, with Holm over the 18 analogous margins (the feedback and diversity margins do not involve the comparator and are unchanged). For reconstruction both checks agree with the main analysis. For detection the selection-aware intervals include zero at every budget, whereas the MI-greedy margins exclude it at 10–20%; MI-greedy has the lowest detection recall in the pool at 5–20%.

Table 13: Stage-2 primary margins against the feasible design with selection-aware 95% bootstrap intervals, and against MI-greedy on every slide (unadjusted 95% CI resampling studies, unadjusted and Holm-adjusted sign-flip p). Post hoc.

	− feasible design		− MI-greedy on every slide		
Budget	Δ	selection-aware 95% CI	Δ [95% CI]	p	p_H
Reconstruction (PCC): residual policy					
0.5%	−0.092	[−0.132, −0.020]	−0.055 [−0.094, −0.001]	0.057	0.424
1%	−0.057	[−0.087, −0.013]	−0.045 [−0.072, −0.018]	0.008	0.090
2%	−0.025	[−0.047, −0.011]	−0.033 [−0.047, −0.017]	<0.001	0.014
5%	−0.016	[−0.025, −0.008]	−0.016 [−0.025, −0.009]	<0.001	0.003
10%	−0.007	[−0.011, −0.003]	−0.007 [−0.011, −0.003]	<0.001	0.011
20%	−0.002	[−0.004, −0.000]	−0.002 [−0.004, −0.001]	0.010	0.107
Detection (ROI recall): local-variance policy					
2%	+0.008	[−0.002, +0.013]	+0.009 [+0.003, +0.015]	0.017	0.168
5%	+0.006	[−0.004, +0.022]	+0.023 [+0.010, +0.037]	0.005	0.063
10%	+0.017	[−0.011, +0.041]	+0.043 [+0.020, +0.067]	0.003	0.043
20%	+0.026	[−0.018, +0.085]	+0.078 [+0.036, +0.122]	0.003	0.043

G Stage-2 secondary family

Table 14: Stage-2 secondary margins, 22 slides. Unadjusted 95% CIs resampling studies; p_H Holm-adjusted over the secondary family.

Margin	Budget	Δ [95% CI]	p_H
residual − residual, perm. (PCC)	0.5%	+0.031 [+0.010, +0.057]	0.056
residual − hindsight benchmark (PCC)	0.5%	−0.113 [−0.146, −0.084]	<0.001
residual − residual, perm. (PCC)	1%	+0.039 [+0.021, +0.060]	0.002
residual − hindsight benchmark (PCC)	1%	−0.072 [−0.099, −0.051]	<0.001
residual − residual, perm. (PCC)	2%	+0.030 [+0.014, +0.049]	0.014
residual − hindsight benchmark (PCC)	2%	−0.041 [−0.056, −0.029]	<0.001
local-var. − local-var., constant (recall)	2%	+0.004 [−0.001, +0.010]	0.518
local-var. − feasible (PCC)	2%	−0.119 [−0.156, −0.087]	<0.001
residual − residual, perm. (PCC)	5%	+0.009 [+0.001, +0.018]	0.238
residual − hindsight benchmark (PCC)	5%	−0.018 [−0.026, −0.011]	<0.001
local-var. − local-var., constant (recall)	5%	+0.019 [+0.006, +0.032]	0.083
local-var. − feasible (PCC)	5%	−0.146 [−0.178, −0.117]	<0.001
residual − residual, perm. (PCC)	10%	+0.001 [−0.002, +0.004]	1.000
residual − hindsight benchmark (PCC)	10%	−0.008 [−0.012, −0.005]	<0.001
local-var. − local-var., constant (recall)	10%	+0.034 [+0.011, +0.057]	0.076
local-var. − feasible (PCC)	10%	−0.150 [−0.185, −0.120]	<0.001
residual − residual, perm. (PCC)	20%	−0.001 [−0.003, +0.002]	1.000
residual − hindsight benchmark (PCC)	20%	−0.005 [−0.007, −0.003]	0.003
local-var. − local-var., constant (recall)	20%	+0.074 [+0.032, +0.118]	0.038
local-var. − feasible (PCC)	20%	−0.163 [−0.215, −0.123]	<0.001

H Spiked positive control

Table 15: Semi-synthetic control on the 22 slides: residual policy (and its permuted-feedback version) minus the feasible design, by spike amplitude and budget (PCC; unadjusted 95% CIs resampling studies). The registered success criterion was a positive margin with a CI excluding zero at $a_{\text{spike}} = 2$.

a_{spike}	Budget	spiked genes in top 50	residual − feasible	residual, perm. − feasible	residual − residual, perm.
0	5%	0	-0.015 $[-0.023, -0.008]$	-0.026 $[-0.034, -0.018]$	$+0.010$ $[+0.002, +0.019]$
0	10%	0	-0.007 $[-0.011, -0.003]$	-0.008 $[-0.012, -0.003]$	$+0.001$ $[-0.002, +0.004]$
1	5%	20	-0.022 $[-0.031, -0.013]$	-0.030 $[-0.038, -0.023]$	$+0.007$ $[-0.001, +0.015]$
1	10%	20	-0.010 $[-0.017, -0.004]$	-0.013 $[-0.019, -0.007]$	$+0.003$ $[-0.002, +0.007]$
2	5%	20	-0.021 $[-0.030, -0.012]$	-0.034 $[-0.043, -0.028]$	$+0.013$ $[+0.004, +0.023]$
2	10%	20	-0.011 $[-0.019, -0.004]$	-0.018 $[-0.026, -0.010]$	$+0.007$ $[-0.000, +0.014]$

I Stage-3 absolute values, secondary family, and permutation seeds

Table 16: Stage-3 mean scores over 22 slides: the base design, the hybrids with informative and permuted feedback (two seeds averaged), and the feedback policy run alone for the whole budget (residual policy for reconstruction, local-variance for detection; stage-2 runs). Base designs as in Table 5.

		50% hybrid		80% hybrid		
Budget	base design	informative	permuted	informative	permuted	feedback policy alone
Reconstruction (PCC)						
0.5%	0.449	0.413	0.412	0.441	0.437	0.357
1%	0.534	0.524	0.498	0.525	0.517	0.477
2%	0.599	0.600	0.584	0.605	0.599	0.566
5%	0.659	0.656	0.654	0.658	0.657	0.643
10%	0.685	0.685	0.686	0.686	0.687	0.678
20%	0.707	0.705	0.708	0.706	0.708	0.705
Detection (ROI recall)						
2%	0.023	0.026	0.019	0.024	0.022	0.027
5%	0.064	0.065	0.051	0.063	0.060	0.070
10%	0.127	0.133	0.110	0.128	0.121	0.137
20%	0.252	0.252	0.217	0.249	0.232	0.278

Table 17: Stage-3 secondary margins, 22 slides. Unadjusted 95% CIs resampling studies; p_H Holm-adjusted over the 24 secondary margins. Permuted values are averaged over two seeds.

Margin	Budget	Δ [95% CI]	p_H
reconstruction, 50% hybrid − permuted	0.5%	+0.0015 [−0.0152, +0.0155]	1.000
reconstruction, 80% hybrid − permuted	0.5%	+0.0040 [−0.0012, +0.0100]	1.000
reconstruction, 50% hybrid − permuted	1%	+0.0262 [+0.0065, +0.0540]	0.209
reconstruction, 80% hybrid − permuted	1%	+0.0078 [−0.0002, +0.0161]	0.749
reconstruction, 50% hybrid − permuted	2%	+0.0156 [+0.0074, +0.0254]	0.027
detection, 50% hybrid − permuted	2%	+0.0066 [+0.0039, +0.0093]	0.004
reconstruction, 80% hybrid − permuted	2%	+0.0067 [+0.0040, +0.0094]	0.016
detection, 80% hybrid − permuted	2%	+0.0016 [+0.0002, +0.0028]	0.477
reconstruction, 50% hybrid − permuted	5%	+0.0024 [−0.0013, +0.0062]	1.000
detection, 50% hybrid − permuted	5%	+0.0141 [+0.0059, +0.0215]	0.113
reconstruction, 80% hybrid − permuted	5%	+0.0012 [−0.0012, +0.0039]	1.000
detection, 80% hybrid − permuted	5%	+0.0037 [+0.0011, +0.0059]	0.277
reconstruction, 50% hybrid − permuted	10%	−0.0008 [−0.0027, +0.0010]	1.000
detection, 50% hybrid − permuted	10%	+0.0227 [+0.0077, +0.0352]	0.209
reconstruction, 80% hybrid − permuted	10%	−0.0018 [−0.0031, −0.0006]	0.124
detection, 80% hybrid − permuted	10%	+0.0074 [−0.0004, +0.0139]	0.749
reconstruction, 50% hybrid − permuted	20%	−0.0025 [−0.0044, −0.0006]	0.277
detection, 50% hybrid − permuted	20%	+0.0349 [+0.0097, +0.0570]	0.277
reconstruction, 80% hybrid − permuted	20%	−0.0019 [−0.0039, −0.0002]	0.477
detection, 80% hybrid − permuted	20%	+0.0169 [+0.0041, +0.0291]	0.277
rounds of 5: local-variance − variance-greedy	10%	+0.0233 [−0.0001, +0.0463]	0.749
rounds of 5: local-variance − permuted	10%	+0.0497 [+0.0216, +0.0747]	0.113
rounds of 5: local-variance − one-shot	20%	+0.0352 [−0.0168, +0.0952]	1.000
rounds of 5: local-variance − permuted	20%	+0.0839 [+0.0247, +0.1405]	0.240

Table 18: Stage-2 feedback ablations repeated over permutation seeds, 22 slides (descriptive). Seed 2021 is the registered stage-2 run. SD is across the seed means; the seed-averaged margin uses the mean of the permuted runs, with an unadjusted 95% CI resampling studies.

	mean margin by permutation seed						
Budget	2021 (stage 2)	1	2	3	4	SD	seed-averaged Δ [95% CI]
Reconstruction (PCC): residual − permuted							
0.5%	+0.031	+0.024	+0.016	–	–	0.007	+0.023 [+0.008, +0.042]
1%	+0.039	+0.040	+0.030	–	–	0.006	+0.036 [+0.020, +0.055]
2%	+0.030	+0.023	+0.025	–	–	0.004	+0.026 [+0.013, +0.041]
5%	+0.009	+0.010	+0.012	–	–	0.002	+0.010 [+0.002, +0.019]
10%	+0.001	0.000	+0.003	–	–	0.002	+0.001 [−0.002, +0.005]
20%	−0.001	−0.001	+0.001	–	–	0.001	0.000 [−0.002, +0.001]
Detection (ROI recall): local-variance − permuted							
2%	+0.005	+0.003	+0.003	+0.006	+0.007	0.002	+0.005 [−0.000, +0.010]
5%	+0.017	+0.011	+0.016	+0.020	+0.022	0.004	+0.017 [+0.005, +0.030]
10%	+0.029	+0.025	+0.034	+0.035	+0.042	0.006	+0.033 [+0.012, +0.056]
20%	+0.063	+0.075	+0.083	+0.060	+0.090	0.013	+0.074 [+0.031, +0.118]

J Gene-set sensitivity

Table 19: Stage-1 local-variance minus the hindsight benchmark of the coverage designs, reconstruction (PCC), under three gene sets, 22 slides, unadjusted 95% CIs resampling studies (secondary).

Budget	per-slide top-50 HVG	reference-subset top-50 HVG	all filtered genes
2%	−0.136 [−0.171, −0.106]	−0.134 [−0.168, −0.103]	−0.032 [−0.047, −0.021]
5%	−0.148 [−0.180, −0.119]	−0.148 [−0.182, −0.118]	−0.052 [−0.068, −0.040]
10%	−0.152 [−0.186, −0.122]	−0.153 [−0.189, −0.122]	−0.068 [−0.087, −0.052]
20%	−0.165 [−0.217, −0.125]	−0.166 [−0.217, −0.126]	−0.079 [−0.101, −0.061]

K Platform and tissue splits

The platform split is exploratory and the per-tissue means are descriptive; both use stage-1 margins.

Table 20: Selected stage-1 margins by platform (unadjusted study-level sign-flip p) and by tissue (mean).

	HD (n=7, 5 cl.)	Visium (n=15, 13 cl.)	prostate (n=6)	kidney (n=4)	lung (n=6)	colon (n=6)
Recon., error-red. (1%)	−0.113 (0.062)	−0.068 (<0.001)	−0.078	−0.085	−0.081	−0.088
Recon., local-var. (10%)	−0.103 (0.062)	−0.174 (<0.001)	−0.115	−0.171	−0.141	−0.185
Detect., local-var. (10%)	+0.048 (0.062)	+0.008 (0.426)	−0.001	+0.022	+0.036	+0.027
Detect., local-var. (20%)	+0.099 (0.062)	+0.015 (0.469)	+0.015	+0.024	+0.052	+0.070
Detect., rounds of 5 (10%)	+0.049 (0.250)	+0.026 (0.044)	+0.016	+0.012	+0.046	+0.053

L Native SOFisher run

We ran SOFisher from the authors' repository on its bundled mouse cortex data for 5 million steps (seed 222) with the repository's cortex script. The only change was checkpoint and resume support, so the run could continue after interruptions. Figure 4 shows the evaluation reward, which counts captured target regions. Its median was −238 over the first 5% of training and +147 over the last 10% (best 514). Rare evaluation episodes have rewards near -1.7×10^5, so the figure shows rolling medians.

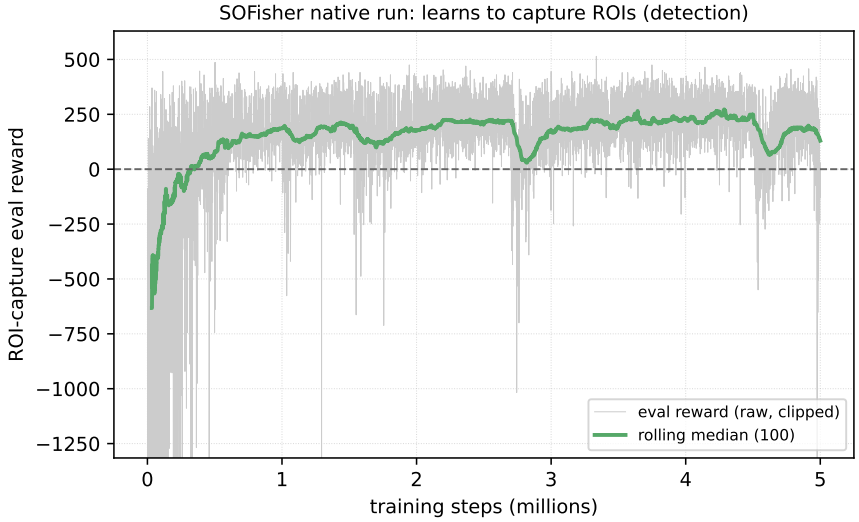


Figure 4: SOFisher evaluation reward during training (rolling median), native run.

M Post hoc robustness checks

Three checks on existing results (post hoc, descriptive). Random selection of n of N regions captures on average a fraction n/N of any fixed ROI set; Table 21 gives each design's recall above this expectation. Table 22 sets the registered bootstrap intervals beside intervals from inverting the exact sign-flip test; the two disagree about excluding zero for three of 38 margins, including the feedback margins at 5% and 10%. Table 23 repeats the primary analyses without TENX174, the slide used to set the imputer's hyperparameters; no conclusion changes.

Table 21: Detection recall minus the recall expected of random selection, 22 slides; 95% CI from inverting the study-level sign-flip test.

	expected recall	recall minus expected random recall, Δ [95% CI]			
Budget	of random selection	local-variance	feasible design	diversity sampler	MI-greedy
2%	0.020	$+0.007$ $[+0.000, +0.013]$	-0.002 $[-0.005, +0.002]$	$+0.006$ $[+0.001, +0.012]$	-0.002 $[-0.005, +0.000]$
5%	0.050	$+0.020$ $[+0.005, +0.034]$	$+0.014$ $[+0.005, +0.024]$	$+0.017$ $[+0.005, +0.032]$	-0.004 $[-0.008, +0.000]$
10%	0.100	$+0.037$ $[+0.011, +0.061]$	$+0.020$ $[-0.000, +0.045]$	$+0.036$ $[+0.016, +0.060]$	-0.006 $[-0.011, -0.002]$
20%	0.200	$+0.078$ $[+0.029, +0.125]$	$+0.052$ $[+0.007, +0.104]$	$+0.066$ $[+0.036, +0.103]$	0.000 $[-0.008, +0.010]$

Table 22: Registered primary margins with the registered bootstrap interval and the sign-flip inversion interval.

Margin	Budget	Δ [registered bootstrap 95% CI]	sign-flip inversion 95% CI
Stage 2 primary family			
residual − feasible (PCC)	0.5%	-0.092 $[-0.132, -0.054]$	$[-0.135, -0.049]$
residual − feasible (PCC)	1%	-0.057 $[-0.089, -0.028]$	$[-0.089, -0.026]$
residual − feasible (PCC)	2%	-0.025 $[-0.040, -0.009]$	$[-0.042, -0.007]$
residual − feasible (PCC)	5%	-0.016 $[-0.025, -0.009]$	$[-0.025, -0.007]$
residual − feasible (PCC)	10%	-0.007 $[-0.011, -0.003]$	$[-0.011, -0.003]$
residual − feasible (PCC)	20%	-0.002 $[-0.004, -0.001]$	$[-0.004, -0.001]$
local-variance − permuted (recall)	2%	$+0.005$ $[-0.001, +0.011]$	$[-0.002, +0.012]$
local-variance − feasible (recall)	2%	$+0.008$ $[+0.002, +0.014]$	$[+0.001, +0.015]$
local-variance − diversity (recall)	2%	$+0.001$ $[-0.004, +0.006]$	$[-0.005, +0.006]$
local-variance − permuted (recall)	5%	$+0.017$ $[+0.002, +0.033]$	$[-0.001, +0.034]$
local-variance − feasible (recall)	5%	$+0.006$ $[-0.005, +0.016]$	$[-0.006, +0.017]$
local-variance − diversity (recall)	5%	$+0.003$ $[-0.007, +0.012]$	$[-0.009, +0.013]$
local-variance − permuted (recall)	10%	$+0.030$ $[+0.004, +0.056]$	$[-0.000, +0.058]$
local-variance − feasible (recall)	10%	$+0.017$ $[-0.007, +0.041]$	$[-0.010, +0.042]$
local-variance − diversity (recall)	10%	$+0.001$ $[-0.017, +0.020]$	$[-0.020, +0.021]$
local-variance − permuted (recall)	20%	$+0.063$ $[+0.016, +0.111]$	$[+0.008, +0.114]$
local-variance − feasible (recall)	20%	$+0.026$ $[-0.018, +0.076]$	$[-0.025, +0.076]$
local-variance − diversity (recall)	20%	$+0.012$ $[-0.021, +0.048]$	$[-0.025, +0.049]$
Stage 3 primary family			
reconstruction 50% hybrid − k-means	0.5%	-0.036 $[-0.057, -0.018]$	$[-0.059, -0.015]$
reconstruction 80% hybrid − k-means	0.5%	-0.009 $[-0.021, +0.003]$	$[-0.023, +0.005]$
reconstruction 50% hybrid − k-means	1%	-0.010 $[-0.020, +0.002]$	$[-0.021, +0.003]$
reconstruction 80% hybrid − k-means	1%	-0.009 $[-0.019, -0.001]$	$[-0.020, +0.001]$
reconstruction 50% hybrid − MI-greedy	2%	$+0.001$ $[-0.007, +0.011]$	$[-0.008, +0.012]$
reconstruction 80% hybrid − MI-greedy	2%	$+0.007$ $[+0.002, +0.013]$	$[+0.001, +0.013]$
reconstruction 50% hybrid − MI-greedy	5%	-0.003 $[-0.007, +0.002]$	$[-0.008, +0.003]$
reconstruction 80% hybrid − MI-greedy	5%	-0.001 $[-0.004, +0.002]$	$[-0.004, +0.003]$
reconstruction 50% hybrid − MI-greedy	10%	0.000 $[-0.002, +0.002]$	$[-0.002, +0.003]$
reconstruction 80% hybrid − MI-greedy	10%	$+0.001$ $[-0.001, +0.002]$	$[-0.001, +0.002]$
reconstruction 50% hybrid − MI-greedy	20%	-0.002 $[-0.004, +0.000]$	$[-0.004, +0.000]$
reconstruction 80% hybrid − MI-greedy	20%	-0.001 $[-0.003, +0.001]$	$[-0.003, +0.001]$
detection 50% hybrid − variance-greedy	2%	$+0.003$ $[+0.000, +0.005]$	$[+0.000, +0.005]$
detection 80% hybrid − variance-greedy	2%	0.000 $[-0.001, +0.002]$	$[-0.001, +0.002]$
detection 50% hybrid − variance-greedy	5%	$+0.002$ $[-0.007, +0.010]$	$[-0.008, +0.011]$
detection 80% hybrid − variance-greedy	5%	0.000 $[-0.003, +0.002]$	$[-0.004, +0.003]$
detection 50% hybrid − variance-greedy	10%	$+0.006$ $[-0.005, +0.018]$	$[-0.007, +0.018]$
detection 80% hybrid − variance-greedy	10%	$+0.001$ $[-0.005, +0.007]$	$[-0.006, +0.007]$
detection 50% hybrid − one-shot	20%	0.000 $[-0.033, +0.034]$	$[-0.038, +0.036]$
detection 80% hybrid − one-shot	20%	-0.003 $[-0.019, +0.014]$	$[-0.023, +0.015]$

Table 23: Registered primary margins with and without the development slide TENX174; p_H Holm-adjusted within each family over 17 studies.

		all 22 slides	without TENX174 (21 slides, 17 studies)	
Margin	Budget	Δ	Δ [95% CI]	p_H
Stage 2 primary family				
residual − feasible (PCC)	0.5%	−0.092	−0.087 [−0.128, −0.049]	0.005
residual − feasible (PCC)	1%	−0.057	−0.055 [−0.088, −0.026]	0.018
residual − feasible (PCC)	2%	−0.025	−0.024 [−0.041, −0.008]	0.134
residual − feasible (PCC)	5%	−0.016	−0.017 [−0.026, −0.009]	0.003
residual − feasible (PCC)	10%	−0.007	−0.007 [−0.011, −0.004]	0.013
residual − feasible (PCC)	20%	−0.002	−0.002 [−0.004, −0.001]	0.113
local-variance − permuted (recall)	2%	+0.005	+0.005 [−0.001, +0.012]	1.000
local-variance − feasible (recall)	2%	+0.008	+0.007 [+0.001, +0.012]	0.370
local-variance − diversity (recall)	2%	+0.001	+0.001 [−0.004, +0.006]	1.000
local-variance − permuted (recall)	5%	+0.017	+0.016 [+0.001, +0.033]	0.805
local-variance − feasible (recall)	5%	+0.006	+0.006 [−0.005, +0.017]	1.000
local-variance − diversity (recall)	5%	+0.003	+0.004 [−0.006, +0.014]	1.000
local-variance − permuted (recall)	10%	+0.030	+0.028 [+0.002, +0.056]	0.769
local-variance − feasible (recall)	10%	+0.017	+0.016 [−0.009, +0.042]	1.000
local-variance − diversity (recall)	10%	+0.001	+0.002 [−0.018, +0.022]	1.000
local-variance − permuted (recall)	20%	+0.063	+0.061 [+0.010, +0.112]	0.467
local-variance − feasible (recall)	20%	+0.026	+0.043 [−0.003, +0.093]	0.899
local-variance − diversity (recall)	20%	+0.012	+0.013 [−0.022, +0.053]	1.000
Stage 3 primary family				
reconstruction 50% hybrid − k-means	0.5%	−0.036	−0.037 [−0.059, −0.018]	0.011
reconstruction 80% hybrid − k-means	0.5%	−0.009	−0.009 [−0.022, +0.004]	1.000
reconstruction 50% hybrid − k-means	1%	−0.010	−0.009 [−0.020, +0.003]	1.000
reconstruction 80% hybrid − k-means	1%	−0.009	−0.011 [−0.020, −0.003]	0.503
reconstruction 50% hybrid − MI-greedy	2%	+0.001	+0.001 [−0.007, +0.012]	1.000
reconstruction 80% hybrid − MI-greedy	2%	+0.007	+0.007 [+0.002, +0.014]	0.327
reconstruction 50% hybrid − MI-greedy	5%	−0.003	−0.003 [−0.008, +0.002]	1.000
reconstruction 80% hybrid − MI-greedy	5%	−0.001	−0.001 [−0.004, +0.003]	1.000
reconstruction 50% hybrid − MI-greedy	10%	0.000	0.000 [−0.002, +0.002]	1.000
reconstruction 80% hybrid − MI-greedy	10%	+0.001	+0.001 [−0.001, +0.002]	1.000
reconstruction 50% hybrid − MI-greedy	20%	−0.002	−0.002 [−0.004, +0.000]	1.000
reconstruction 80% hybrid − MI-greedy	20%	−0.001	−0.001 [−0.003, +0.001]	1.000
detection 50% hybrid − variance-greedy	2%	+0.003	+0.002 [−0.000, +0.005]	1.000
detection 80% hybrid − variance-greedy	2%	0.000	0.000 [−0.001, +0.002]	1.000
detection 50% hybrid − variance-greedy	5%	+0.002	+0.002 [−0.007, +0.010]	1.000
detection 80% hybrid − variance-greedy	5%	0.000	0.000 [−0.004, +0.003]	1.000
detection 50% hybrid − variance-greedy	10%	+0.006	+0.006 [−0.006, +0.018]	1.000
detection 80% hybrid − variance-greedy	10%	+0.001	+0.002 [−0.005, +0.008]	1.000
detection 50% hybrid − one-shot	20%	0.000	+0.005 [−0.026, +0.040]	1.000
detection 80% hybrid − one-shot	20%	−0.003	−0.001 [−0.017, +0.017]	1.000